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Tertiary paleomagnetic data from northwestern Yunnan, China: further evidence for large clockwise rotation of the Indochina block and its tectonic implications

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Abstract

Eocene red sandstones were collected at 15 sites for paleomagnetic study from two areas (Lanping and Yunlong) of the western part of Yunnan province, China, in the northern part of Indochina. The high-temperature magnetization component with an unblocking temperature of around 680°C is isolated after stepwise thermal demagnetization. Characteristic directions from nine accepted sites reveal a positive fold test at the 99% confidence limit, indicating that the high-temperature magnetization component is possibly of Eocene age. A tilt-corrected mean direction from nine accepted sites is $D = 266.1°$, $I = -39.8°$ with $\alpha_{95} = 11.2°$, corresponding to a paleopole at 14.5°N, 169.7°E with $A_{95} = 10.9°$. A comparison with previously reported Tertiary paleomagnetic data indicates that the Lanping basin of the northern part of the Indochina block rotated clockwise through 91° (±15°) with respect to the Yangtze craton. This large magnitude clockwise rotation is probably caused by the superimposition of regional and local rotations. Our previous middle Cretaceous paleomagnetic results from the Yunlong area indicate that about 40° of clockwise rotation out of 90° is explained by tectonic rotation due to a relatively rigid body rotation of the Simao Terrane during indentation of India. The remaining component of rotation is probably ascribed to further Eocene/post-Eocene rotational motion of the tectonically independent small blocks within the Lanping basin associated with fault movement. © 2001 Elsevier Science B.V. All rights reserved.

Keywords: plate collision; deformation; Indochina; Eocene; paleomagnetism

1. Introduction

The India continent collided with the Asian continent about 52 Ma and continued penetration into it [1,2]. The 2500 km northward penetration of India gave rise to tectonic deformation in the Asian continent. Crustal thickening of the Asian lithosphere along with latitude shortening in front

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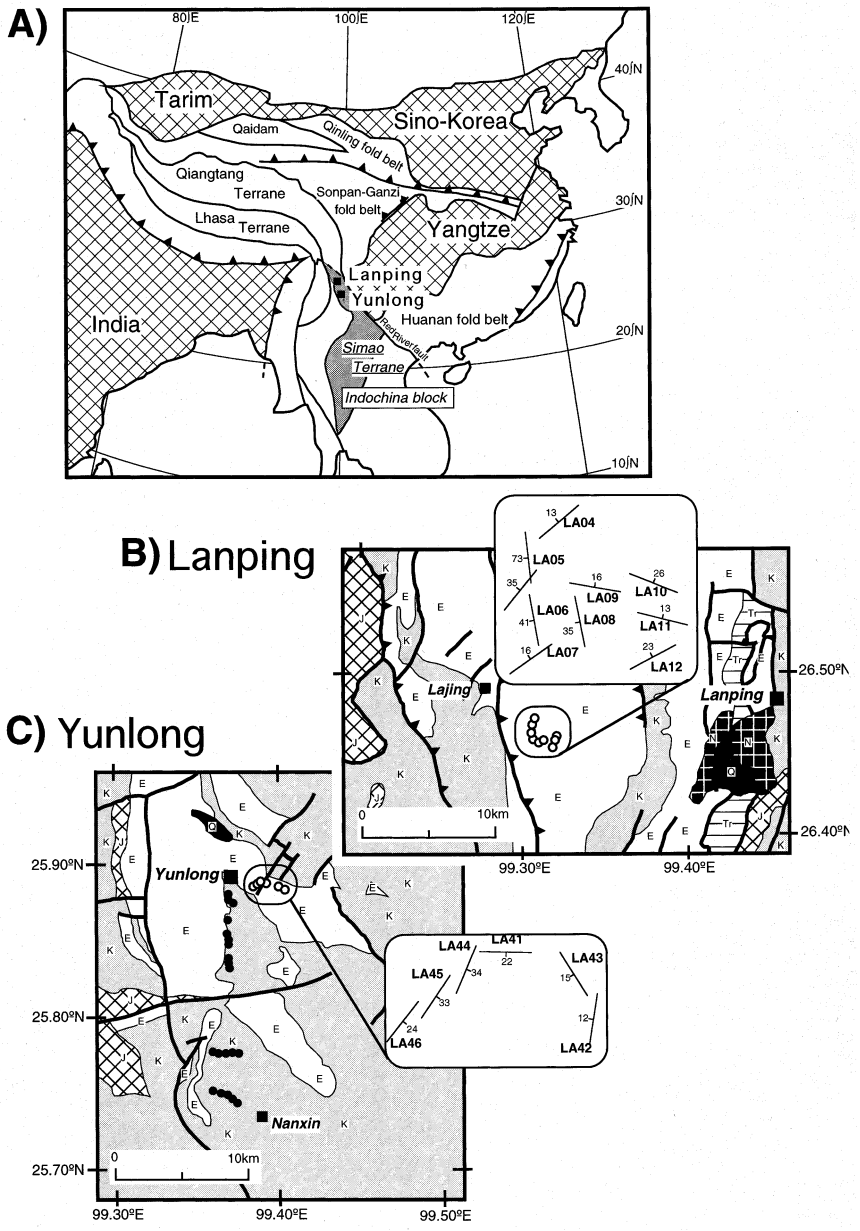


Fig. 1. (A) Sketch tectonic map of Southeast Asia after Sato et al. [8]. Barbed lines are continental thrusts. (B, C) Geologic map of the study area (after [21,22]). Sampling sites are shown by open circles. Closed circles in (C) denote the sampling sites of Cretaceous rocks studied by Sato et al. [8]. Black lines and barbed lines denote faults and thrusts, respectively. Tr, Triassic; J, Jurassic; K, Cretaceous; E, Eocene; M, Miocene; Q, Quaternary. Strike and dip of strata for each sampling site are shown in insets.

of India and tectonic rotation along both sides are anticipated from both laboratory and numerical experiments [3–6].

Southeast Asia consists of three large cratons (the Sino-Korea craton, the Tarim craton and the Yangtze craton) and several accretionary terranes (e.g. the Simao Terrane, the Qiangtang Terrane and the Lhasa Terrane) [7] (Fig. 1A). Be-

cause the Simao Terrane, the northern part of the Indochina block, is situated between the Indian continent and the Yangtze block, the deformation feature of the Asian continent due to the penetration of India is expected to be preserved in this terrane.

Several paleomagnetic results from Cretaceous rocks have been reported from the Simao Terrane in the last decade [8–11]. Paleomagnetic evidence for clockwise rotations that vary between 36° and 112° has been provided from Cretaceous results. Tertiary paleomagnetic data are, however, still scarce from the Simao Terrane. Chen et al. [11] reported Tertiary paleomagnetic data from the Simao Terrane, although their fold test is based at the sample level. In order to define a model for the tectonic evolution of the Simao Terrane during the Cenozoic, additional Tertiary paleomagnetic data are required in this terrane.

We present new paleomagnetic results from Tertiary redbeds in the northern part of the Simao Terrane. The Lanping area and Yunlong area (Fig. 1B,C) were chosen as paleomagnetic study areas. Our paleomagnetic study focuses on Eocene strata, in order to assess the magnitude of deformation of the Simao Terrane in Cenozoic time from paleomagnetic view points.

2. Geologic setting and sampling

The Simao Terrane is located east of Tibet and southwest of the Yangtze craton (Fig. 1A). The Simao Terrane is bounded on the northeast by the NW–SE-trending Red River fault, which marks the boundary of the Yangtze craton and the Indochina block. A large amount of sinistral motion on the Red River fault has been reported from geologic and chronologic data [12–18]. West of the Red River fault zone are tectonic collages of continental fragment [19]. The Simao Terrane is one of the well-defined continental fragments and is bounded on the west by the Changning–Menglian suture [20,21]. In the Simao Terrane, Mesozoic to Cenozoic continental sedimentary rocks are widely exposed. This thick sequence overlies Paleozoic to Mesozoic strata collectively reformed to as 'the Cathaysia' [20].

The Lanping basin is composed of Triassic to Eocene redbed strata in the northern part of the Simao Terrane [22]. Cretaceous and Eocene strata form folding with a roughly north–south trending axis [21]. Around the Lanping county (Fig. 1B), the upper Eocene sequence is composed of the Baoxiangsi and Jinsichang Formations. The Baoxiangsi Formation, 962.2 m in thickness, consists of yellow–white quartz–feldspar sandstones and purplish red sandstones, and is unconformably overlain by the Jinsichang Formation. Sporopollen, plant fossils and mammalian elements such as *Eoentelodon likiangensis*, *Lunania* cf. and *Honanodon hebertis* assign a Late Eocene age to the Baoxiangsi Formation.

Yunlong county (25.8°N, 99.4°E) (Fig. 1C), about 60 km south of Lanping county, is situated in the central part of the Lanping basin. Around Yunlong county, the Eocene sequence is composed of three formations [22]: the lower Eocene Mengyejing Formation conformably underlies the middle Eocene Denghei Formation which is unconformably overlain by the upper Eocene Mengla Formation. The Denghei Formation, 1620.4 m in thickness, consists mainly of purplish red sandstones. The occurrence of sporopollen, gastropod charophytes, *Pinnocypris gibba*, *Pinnocypris yunnanesis* and *Limnocythere spinosalata* assigns a Middle Eocene age to the Denghei Formation.

We have collected a total of 122 samples of Eocene red sandstones from these two areas. In the Lanping area, we have sampled red sandstones at nine sites of the upper Eocene Baoxiangsi Formation. Anticline folding in the Lanping area took place in the Miocene, as evidenced by an unconformity between the upper Eocene Jinsichang Formation and the Miocene Shuanghe Formation [22]. In the Yunlong area, we have also collected red sandstones at six sites of the middle Eocene Denghei Formation. Syncline folding in the Yunlong area occurred in the Late Eocene as evidenced by an unconformity between the middle Eocene Denghei Formation and the upper Eocene Mengla Formation [22]. Five to 10 hand samples, oriented with a magnetic compass, were distributed over a distance up to 15 m in a site. Localities of the sampling sites were

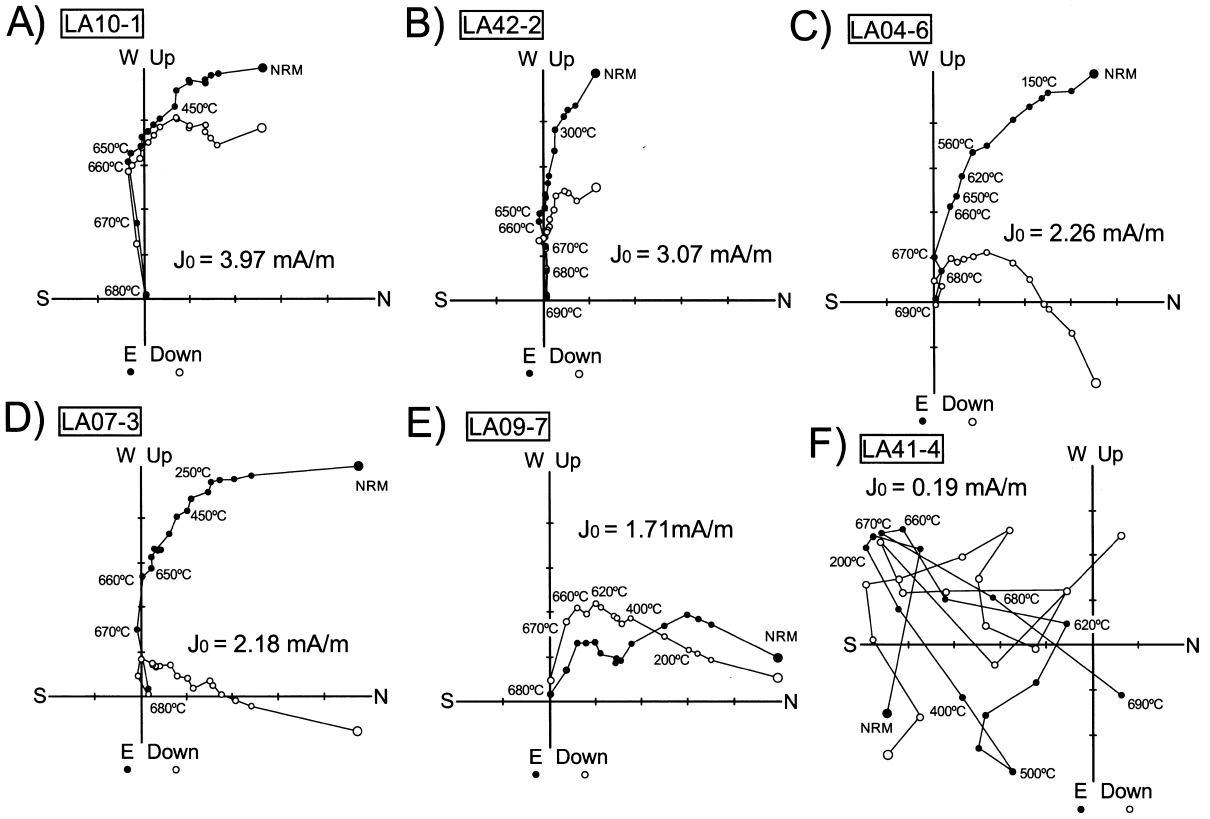


Fig. 2. Orthogonal projections of magnetization vector end-points for the representative specimens from the upper Eocene Baoxiangsi Formation (A, C, D, E) and the middle Eocene Denghei Formation (B, F) during thermal demagnetization experiments. Solid symbols for horizontal projection, open symbols for vertical. Tectonic tilt correction has not been applied.

determined by a portable navigational instrument (Sony Pyxis).

The present declination value at each sampling locality was estimated from the calculation using the International Geomagnetic Reference Field [23].

3. Paleomagnetism

In the laboratory, one or more specimens, 25 mm in diameter and about 23 mm in height, were prepared from each sample. Remanent magnetization of each specimen was measured using a 2G Enterprises cryogenic magnetometer. A stepwise thermal demagnetization was carried out up to 690°C with a laboratory-built furnace or a Natsuhara TDS-1 thermal demagnetizer; the residual magnetic field in these furnaces was less than 5 nT. The magnetic behavior for each specimen during demagnetization was plotted on a Zijderveld diagram [24]. Principal component analysis [25] was used to determine the component directions of magnetization and site-mean direction was calculated using Fisherian statistics [26].

Progressive acquisition of isothermal remanent magnetization (IRM) up to 2.7 T and a composite IRM [27] were performed on a 2G pulse magnetizer in order to identify ferromagnetic minerals. The anisotropy of magnetic susceptibility (AMS) of some specimens was measured using KLY-3 susceptibility bridge apparatus.

3.1. Demagnetization

Specimens of the upper Eocene Baoxiangsi For-

mation from Lanping have a natural remanent magnetization (NRM) intensity between 4.0 and 0.5 mA/m. All specimens have multi-component magnetizations; generally two-component magnetization appears during progressive thermal demagnetization. A low-temperature magnetization component is removed by 500°C. Some specimens preserve it up to 660°C (Fig. 2E). After removal of the low-temperature component, remanent magnetization decay towards the origin (Fig. 2A,C). The unblocking temperature of a high-temperature component ranges between about 670 and 690°C. The high unblocking temperature above 670°C indicates that this remanent magnetization is carried by hematite.

Specimens of the middle Eocene Denghei Formation from Yunlong have NRM intensities of 3.1–0.2 mA/m. Specimens of site LA42 and LA43, which have NRM intensities of 3.1–1.0 mA/m, show similar thermal demagnetization behavior as specimens of Lanping; the high-temperature component with the unblocking temperature of above 670°C is resolved (Fig. 2B). In contrast, specimens of site LA41, LA44, LA45 and LA46, which have relatively weak NRM intensities of 1.1–0.2 mA/m, show unstable demagnetization behavior during progressive thermal demagnetization (Fig. 2F).

Acquisition of IRM and back-field demagnetization indicate that hematite is a major magnetic

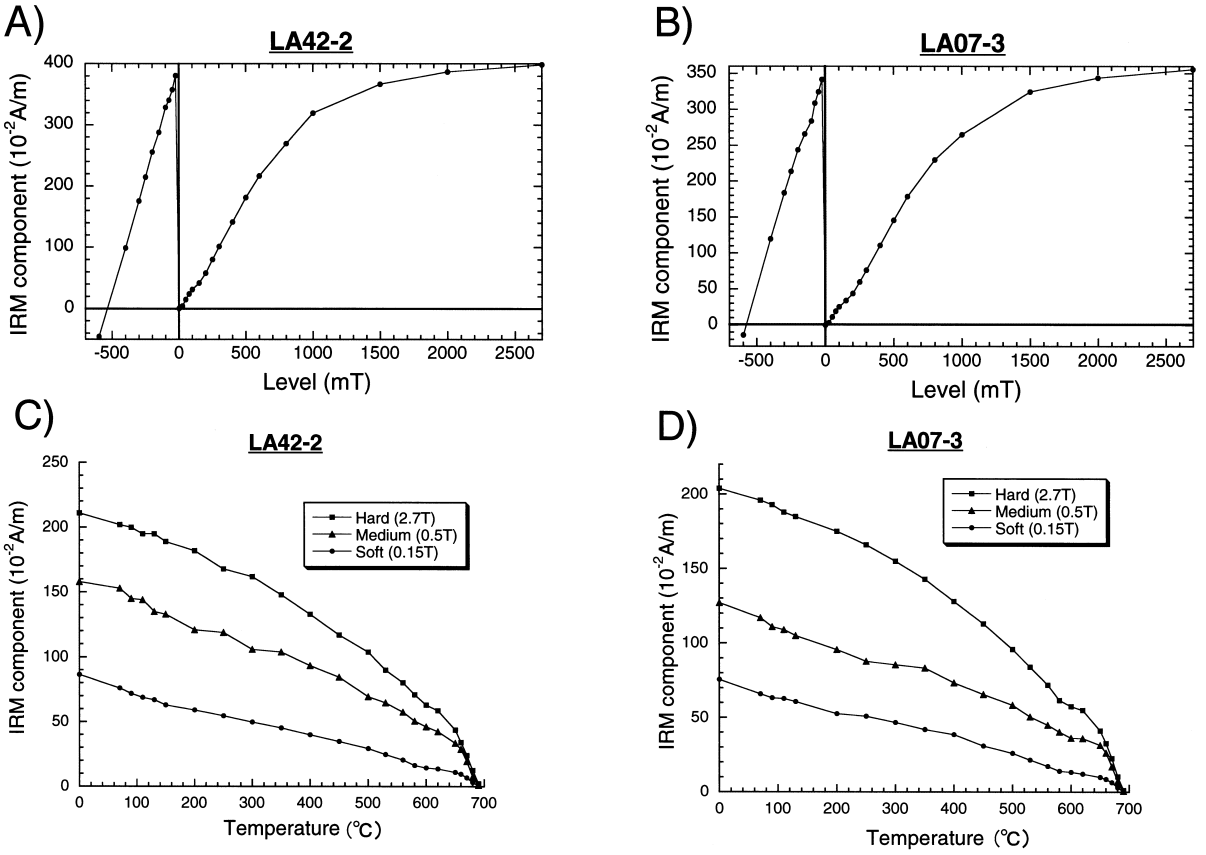


Fig. 3. IRM acquisition curve, back-field demagnetization and thermal demagnetization of IRM for specimens (LA42-2 and LA07-3) of sites LA42 and LA07. (A, B) Specimens do not reach saturation IRM at a field of 2.7 T in dc field. A shoulder at about 200 mT is revealed in both specimens. (C, D) Thermal demagnetization of a three component IRM produced by applying a different dc field (2.7, 0.5 and 0.15 T) to each of three perpendicular axes of a specimen. Thermal demagnetization curves of all the components show the hematite unblocking temperature, whereas the magnetite unblocking temperature of 580°C is revealed in demagnetization curves of the hard component.

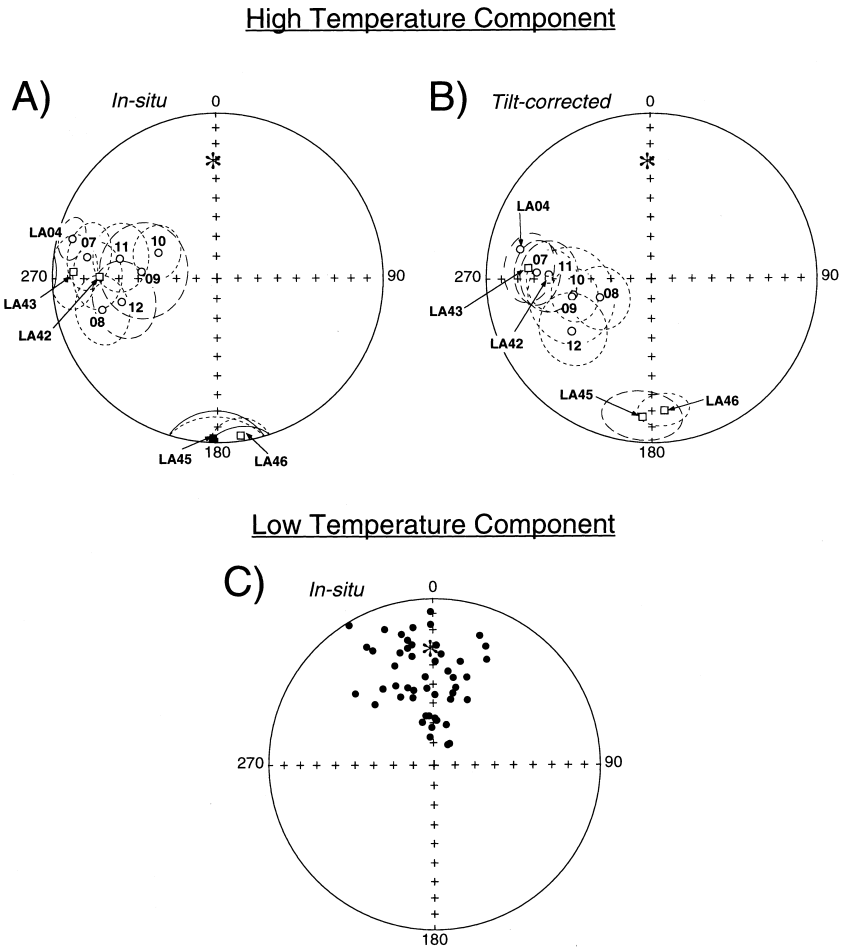


Fig. 4. Equal-area projections of the site mean directions of high-temperature components with circles of 95% confidence limit before (A) and after (B) tilt correction. Circles are the site mean direction from the Lanping area and squares are from the Yunlong area. Numbers by symbols represent site name. (C) Equal-area projections of low-temperature component before tilt correction. The asterisk represents the present direction of the Earth's magnetic field. Solid symbols refer to the lower hemisphere, open symbols refer to the upper hemisphere.

carrier in the redbeds of the Eocene strata in the Lanping and Yunlong sections (Fig. 3); a high saturation field more than 2.7 T and a high coercivity of the remanence of 500–600 mT. This is confirmed by the unblocking temperature of hematite of 680°C in thermal demagnetization of the composite IRM. The presence of a trace amount of magnetite is also noticed from a 'shoulder' at a field of 200 mT in the IRM acquisition curve and an unblocking temperature of around 580°C.

3.2. Paleomagnetic direction

We have resolved two components (low and high temperature) of magnetization from two areas during stepwise thermal demagnetization.

Low-temperature components were isolated from 50 samples (Fig. 4C, Table 1). The mean direction of this component is $D = 354.3°$, $I = 43.9°$, with $\alpha_{95} = 6.1°$ in situ, and $D = 346.5°$, $I = 30.9°$ with $\alpha_{95} = 9.5°$ after tilt correction. The precision parameter k decreases from 11.8 to 5.5

($k_s/k_g = 0.47$) after tilt correction. The mean in situ direction is sub-parallel to the present direction of the Earth's magnetic field ($D = 359.4$, $I = 39.5°$), suggesting that this component is attributed to viscous remanent magnetization acquired in recent times.

High-temperature components were isolated from 65 out of 122 samples in 12 sites. Except for site LA06, paleomagnetic directions of the high-temperature component are well clustered; the precision parameter k ranges from 11.1 to 63.0 (Fig. 4A,B, Table 1). Mean directions of site LA06 show large dispersion ($\alpha_{95} = 29.3°$), so that this site is excluded from later discussion.

The remaining 11 sites are divided into two groups. Nine sites from both Lanping and Yunlong areas form one group. Their paleomagnetic directions are characterized by west-directed declination ($D = 236.5$–$299.8°$). The inclination value ranges from −19.1° (LA04) to −62.5°(LA08). The other group (LA45 and LA46 from Yunlong area) is characterized by south-directed declination with upward shallow inclination.

AMS results suggest that samples from the nine sites with west-directed declination preserve a horizontal structure of the depositional processes. Their magnetic fabrics are characterized by oblate susceptibility ellipsoid. Axial distribution of AMS

Table 1
Paleomagnetic results for redbeds in the Eocene formations of the Lanping basin

Site	Locality		n/N	In-situ		Tilt-corrected		k	α_{95}	Formation
	(°N)	(°E)		dec. (°)	inc. (°)	dec. (°)	inc. (°)		(°)	
High-temperature component:										
Lanping area										
LA04	26.47	99.31	6/8	285.3	−8.7	283.3	−19.1	63.0	8.6	Baoxiangsi Fm.
LA05[a]	26.47	99.30	0/10	–	–	–	–	–	–	Baoxiangsi Fm.
LA06[a]	26.41	99.31	5/8	284.0	−17.5	299.8	−53.3	7.8	29.3	Baoxiangsi Fm.
LA07	26.46	99.31	5/8	279.3	−20.0	273.5	−30.5	31.5	13.8	Baoxiangsi Fm.
LA08	26.46	99.31	4/8	254.8	−27.7	250.4	−62.5	37.4	15.2	Baoxiangsi Fm.
LA09	26.46	99.32	4/8	275.6	−51.6	257.7	−47.8	17.1	22.9	Baoxiangsi Fm.
LA10	26.46	99.32	7/8	293.5	−57.3	258.8	−49.5	24.4	12.5	Baoxiangsi Fm.
LA11	26.46	99.32	6/8	282.1	−38.7	272.0	−37.0	19.4	15.6	Baoxiangsi Fm.
LA12	26.45	99.33	8/9	256.6	−39.8	236.5	−41.5	11.1	17.4	Baoxiangsi Fm.
Formation mean direction			7/9	274.7	−35.6			15.5	15.8	
						263.8	−42.1	19.3	14.1	
Yunlong area										
LA41[a]	25.89	99.39	0/5	–	–	–	–	–	–	Denghei Fm.
LA42	25.89	99.40	4/9	270.9	−26.5	269.7	−38.4	41.7	14.4	Denghei Fm.
LA43	25.89	99.40	7/9	272.7	−13.0	275.8	−25.1	17.4	14.9	Denghei Fm.
LA44[a]	25.89	99.39	0/8	–	–	–	–	–	–	Denghei Fm.
LA45[a]	25.89	99.39	4/8	181.6	0.4	185.2	−16.9	33.4	16.1	Denghei Fm.
LA46[a]	25.89	99.39	5/8	173.2	−3.6	176.5	−20.3	56.7	10.2	Denghei Fm.
Formation mean direction			2/5	271.8	−19.8			–	–	
						273.0	−31.8	–	–	
Overall mean direction			9/15	274.0	−32.0			17.4	12.7	
						266.1	−39.8	22.0	11.2	
Low-temperature component:										
			50/122	354.3	43.9			11.8	6.1	
						346.5	30.9	5.5	9.5	

The magnetization directions of high- and low-temperature components are listed. N and n are numbers of field oriented samples and samples used for statistics, respectively. Dec. and inc. are declination and inclination, respectively; k is the Fisherian precision parameter [26]; α_{95} is the radius of cone of 95% confidence about the mean direction. Eocene mean directions for high-temperature component are calculated from site mean directions, while mean directions for low-temperature component are calculated on the basis of sample directions.

[a]LA05, LA06, LA41 and LA44–46 are not used for calculation of the mean direction for the high-temperature component.

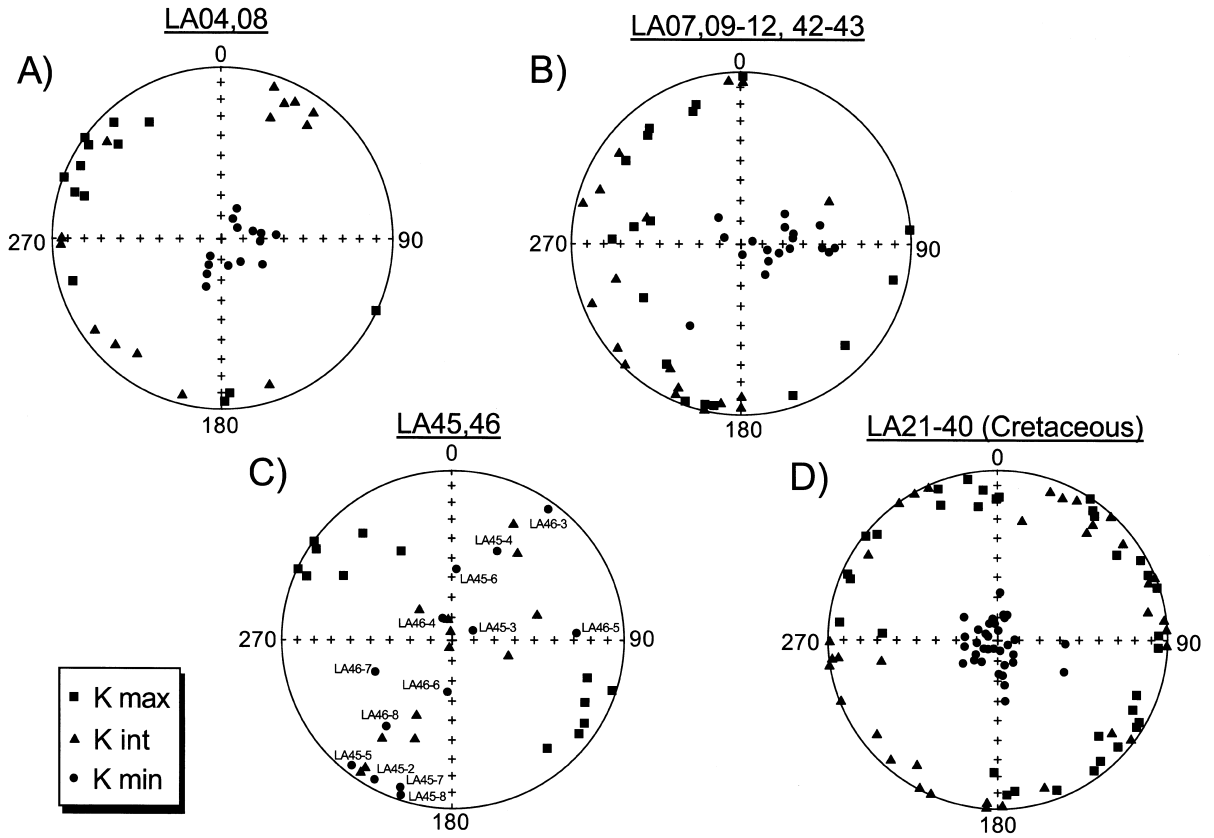


Fig. 5. Equal-area lower hemisphere projections of AMS principle axes after tilt correction. (A) Representative specimens of site LA04 and LA08, (B) site LA07–09, LA42 and LA43, (C) site LA45 and LA46, in the Eocene formations. (D) Site LA20–40 in the Cretaceous formation [8].

from these nine sites after tilt-correction shows that the k_{min} axes are generally vertical, whereas both of the k_{max} and k_{int} are dispersed in the horizontal plane (Fig. 5A,B). These AMS results are comparable with those of Cretaceous redbeds from the Yunlong section [8] (Fig. 5D). A significantly large variance in the inclination of paleomagnetic directions does not appear to be due to AMS effects.

In contrast, no original structure of the depositional process remains in two sites with south-directed declination (LA45 and LA46). The magnetic fabric is characterized by spherical susceptibility ellipsoid. Axial distribution of AMS after tilt-correction is quite different from that of the former results of nine sites: the k_{max} and k_{int} are distributed around a NE–SW trending great circle (Fig. 5C). Since site LA45 and LA46 are situated close to a small fault (Fig. 1C), these sites probably suffered regional strain due to fault movement. The paleomagnetic directions observed in site LA 45 and LA 46 may be due to remagnetizations during fault movement. Consequently, the paleomagnetic data from these two sites are discarded for later discussion.

The mean direction from nine sites with west-directed declination is $D = 274.0°$, $I = -32.0°$ with $\alpha_{95} = 12.7°$ in situ, and $D = 266.1°$, $I = -39.8°$ with $\alpha_{95} = 11.2°$ after tilt correction. The precision parameter k increases from 17.4 to 22.0 ($k_s/k_g = 1.26$) which is, however, insignificant for a positive fold test of McElhinny [28] even at the 95% level. We then apply the McFadden's fold test [29]. The calculated value is $\xi_1 = 5.28$ in situ

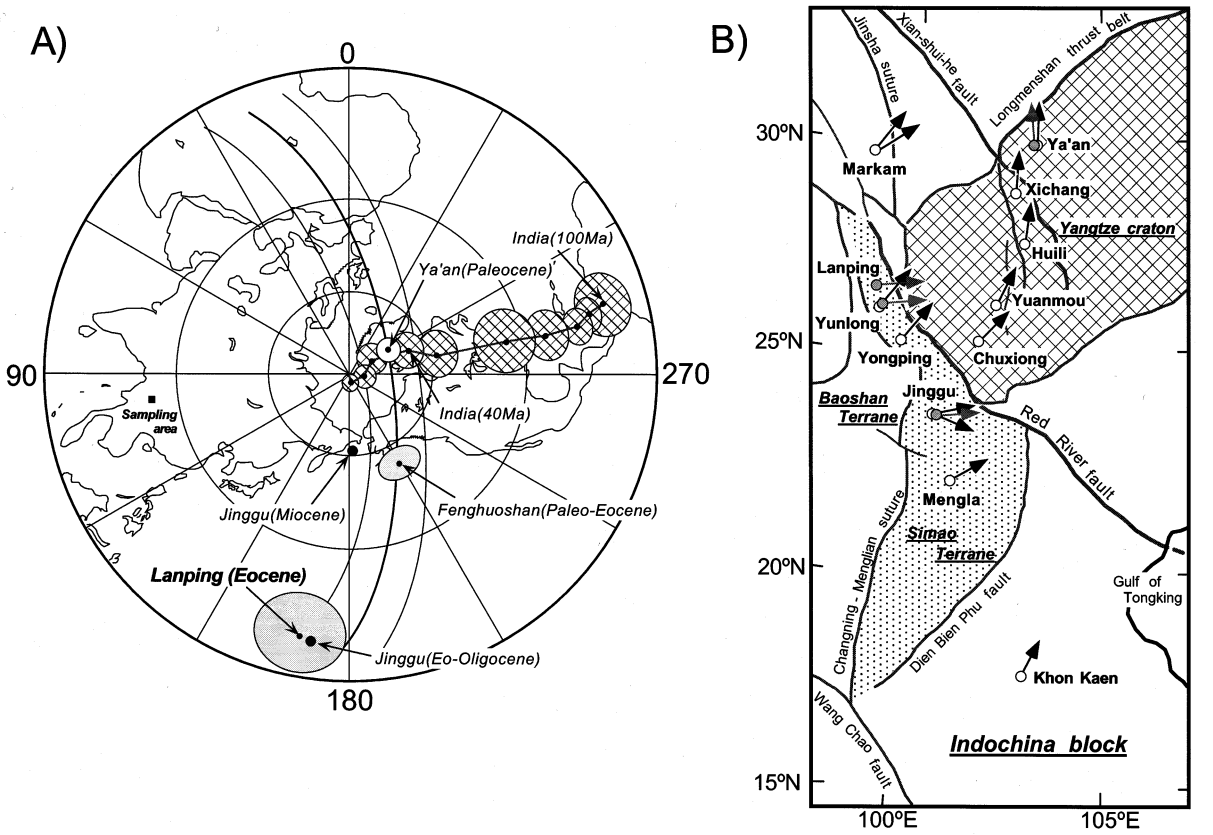


Fig. 6. (A) Tertiary paleomagnetic poles for the Lanping basin and its surrounding regions. The APWP for India since 100 Ma [31] and small circles centered on our study area are also shown. (B) Declinations of the Cretaceous (block) and Tertiary (gray) paleomagnetic directions in the Indochina block and its surrounding regions are shown by arrows. Data are summarized in Table 2.

and $\xi_1 = 1.64$ after tilt correction, while the critical value at the 99% confidence level is $\xi = 4.85$. This fold test is positive at the 99% confidence level. The Eocene formations of the Lanping and Yunlong areas in the Lanping basin preserve magnetizations of a pre-folding origin.

Paleomagnetic poles calculated from the tilt-corrected paleomagnetic directions from sampling sites provides a mean paleomagnetic pole at 14.5°N, 169.7°E with $A_{95} = 10.9°$ (Fig. 7A, Table 2) for the Lanping basin.

4. Discussion

The remanent magnetization of the high-temperature component after tilt correction for the Baoxiangsi Formation in the Lanping area and Denghei Formation in the Yunlong area represents a characteristic paleomagnetic direction for the Lanping basin, probably of Eocene age. This direction reveals a large clockwise deflection in declination from the north. The reliability of primary and at least pre-folding origin for the remanent magnetization is ascertained through successful thermal demagnetization and a positive fold test. It is hard to claim that the west-directed direction with only reversed polarity is due to geomagnetic secular variation or excursion, because this direction is observed from both the upper and middle Eocene strata.

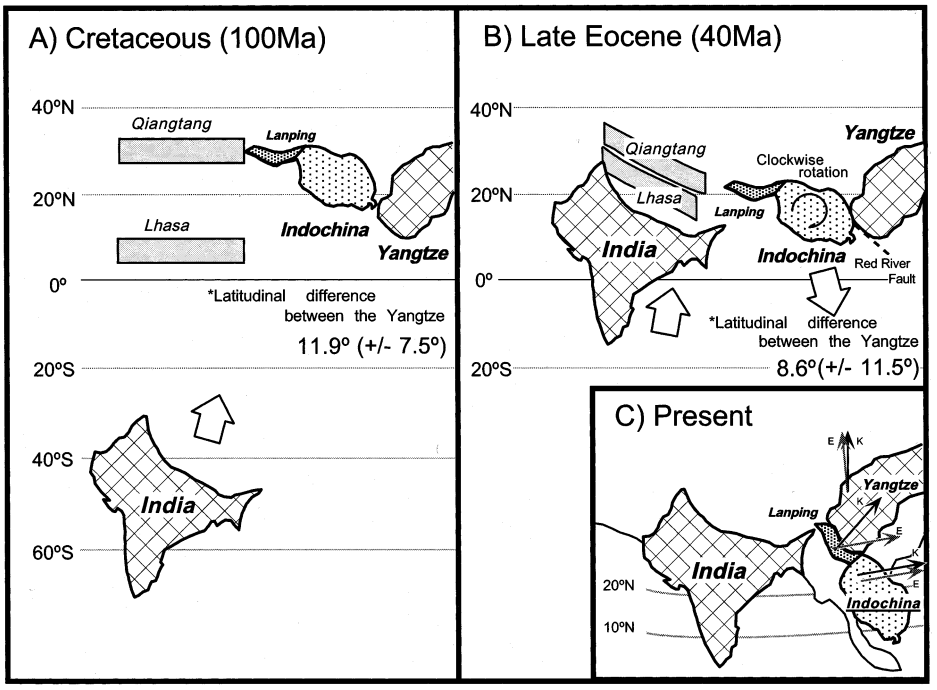


Fig. 7. (A, B) Paleoposition of the Indochina block and surrounding blocks (India, Yangtze craton, Qiangtang Terrane and Lhasa Terrane). Paleoposition of each block is calculated from Cretaceous (A) and Tertiary (B) paleomagnetic directions. Paleoposition of India is estimated from the APWP by Besse and Courtillot [31]. (C) Present position of Indochina block and surrounding blocks. Declination of the Cretaceous (K, black) and Tertiary (E, gray) paleomagnetic directions are also shown by arrows.

4.1. *Clockwise rotation in northern part of the Simao Terrane since the Eocene*

Paleomagnetic data clearly show that the northern part of the Simao Terrane was subjected to large magnitude tectonic rotation. The pole from the Lanping basin can be compared with the Tertiary paleomagnetic pole (76.8°N, 299.9°E) of Ya'an (30°N, 103°E) [30] from the western part of the Yangtze craton (Fig. 6A, Table 2). The Lanping pole is located at a clockwise-rotated position from the Ya'an pole. These two poles are significantly distinguishable from each other. Angular distance between these poles is $84.2 \pm 11.5°$. This deviation indicates that large clockwise rotation of the Lanping basin took place with respect to the western part of the Yangtze craton in the Cenozoic.

The amount of rotation is evaluated by comparison of observed direction and that expected from the Tertiary pole of Ya'an [30] at the Lanping basin. The Lanping basin is rotated clockwise through $90.7 \pm 15.4°$ with respect to the western part of the Yangtze craton. Comparing the Lanping basin with the 40 Ma pole of India [31], the amount of clockwise rotation of the Lanping basin is $89.9 \pm 11.9°$, with respect to India. The Lanping basin is rotated clockwise through about 90° with respect to the surrounding cratons.

This Cenozoic extremely large clockwise rotation of the Lanping basin is ascribed to tectonic rotation due to a collision of India and Asia [8]. A wide area of the Asian continent northeast of India has suffered large clockwise rotation later than Cretaceous. Clockwise deflected declinations have been reported from Cretaceous strata from the eastern Qiangtang Terrane ($D = 40$–$58°$), the northern part of Simao Terrane ($D = 40°$, 42°), and the Chuxiong basin in the southern part of the Yangtze craton ($D = 45°$) [8,9,32,39] (Table 2).

Table 2
Cretaceous and Tertiary paleomagnetic results obtained from the Indochina block and its surrounding regions

Locality			Age	N	Observed direction			VGP			Rotation	Latitudinal difference	relative to	Reference
name	lat. (°N)	long. (°E)			dec. (°)	inc. (°)	α_{95} (°)	lat. (°N)	long. (°E)	A_{95} (°)	(°)	(°)		
Indochina block:														
Simao Terrane														
1 Lanping	26.5	99.3	E	9	266.1	−39.8	11.2	14.5	169.7	10.9	90.7 ± 15.4	8.6 ± 11.5	13	this study
2 Yunlong	25.8	99.4	K2	20	40.2	49.9	3.9	54.6	171.3	4.4	36.3 ± 13.6	11.9 ± 7.5	16	[8]
3 Xiaguan	25.6	100.2	K2	9	6.9	47.7	8.6	83.6	152.7	10.0	2.9 ± 17.7	10.2 ± 9.9	16	[10]
4 Yongping	25.5	99.5	K1	12	42.0	51.1	15.7	50.9	167.3	20.6	38.1 ± 28.3	12.0 ± 9.4*	16	[9]
5 Jinggu	23.4	100.4	K1	–	84.4	39.6	17.8	13.6	171.5	–	80.3 ± 23.7	6.0 ± 17.3	16	[11]
6 Jinggu	23.4	100.5	K2	–	295.8	−36.0	6.3	−13.9	161.3	–	111.7 ± 14.6	3.5 ± 8.8	16	[11]
7 Jinggu	23.5	100.8	E–O	–	84.7	38.9	7.6	13.2	172.2	–	89.1 ± 10.9	11.0 ± 6.8	13	[11]
8 Jinggu	23.5	100.7	M	–	21.1	35.5	7.5	70.0	183.2	–	25.5 ± 10.4	8.6 ± 6.5	13	[11]
9 Jinggu	23.4	100.9	K2	8	79.4	43.3	9.1	18.9	170.0	8.9	75.3 ± 17.6	8.7 ± 9.7	16	[10]
10 Mengla	21.6	100.4	K2	10	60.8	37.8	7.6	33.7	179.3	8.2	56.8 ± 15.7	6.5 ± 8.5	16	[10]
Khorat plateau														
11 Khon Kaen	16.5	103.0	K1	10	28.1	40.5	2.4	62.7	173.3	2.4	23.8 ± 13.2	13.3 ± 6.9	16	[34]
Yangtze craton:														
12 Ya'an	30.0	102.9	Ju-P	26	2.0	34.2	3.6	78.6	273.4	2.9				[30]
13 Ya'an	30.0	103.0	P	7	356.0	31.9	4.1	76.8	299.9	3.6				[30]
14 Xichang	27.9	102.3	K1	36	3.4	41.2	5.3	84.8	245.8	5.7				[36]
15 Huili	26.6	102.3	K2	18	8.1	38.8	6.6	81.5	220.9	7.1				[37]
16 (stable Yangtze craton (mean of 12, 15, 15))								82.2	250.3	7.6				
17 Yuanmou	25.9	101.8	K2	21	26.9	35.6	3.6	64.6	199.6	3.3				[38]
18 Chuxiong	25.0	101.5	K1	21	44.6	41.3	10.7	49.2	182.0	11.4				[39]
Qiangtang Terrane:														
19 Markam	29.7	98.7	K2	11	39.7	51.6	8.5	56.7	172.7	10.6	35.8 ± 18.2	8.4 ± 11.9	16	[33]
20 Markam	29.7	98.4	K1	12	57.7	50.2	10.9	40.6	170.5	13.0	53.8 ± 20.9	9.6 ± 10.4	16	[33]
21 Fenghuoshan	34.5	92.8	P	7	9.5	34.3	8.9	62.6	210.5	3.9/6.8	15.9 ± 11.8	−3.4 ± 7.7	13	[40]

Lat. and Long. are latitude and longitude, respectively. Age: Ju, Upper Jurassic; K1, Lower Cretaceous; K2, Upper Cretaceous; P, Paleocene; E, Eocene; M, Miocene. N is the number of sites used for statistics. Dec. and inc. are declination and inclination, respectively. α_{95} and A_{95} are the radius of cone of 95% confidence about the mean direction and the VGP, respectively. Rotation and latitudinal difference are evaluated by comparison between the observed paleomagnetic declination and that calculated from the western part of the Yangtze craton. Cretaceous paleomagnetic data are compared with Cretaceous paleomagnetic pole of the stable Yangtze craton calculated by Sato et al. [8], whereas the Tertiary paleomagnetic data are compared with the Tertiary paleomagnetic pole of Ya'an [30]. Latitudinal difference of Yongping (*) is evaluated using recalculated inclination (bivariate mean) by Yang et al. [36].

Because laboratory and numerical experiments predict a large magnitude clockwise rotation at regions to the northeast of the Indian craton [3–6], these clockwise deflected paleomagnetic directions of Cretaceous–Eocene are attributed to tectonic rotation due to deformation of the Asian continent by the collision event of Eurasia/India. Our Eocene paleomagnetic data reveal that rotation, at least in some areas of the northern part of the Simao Terrane, continued subsequent to the Eocene and post-Eocene.

4.2. Eocene/post-Eocene rotation of small blocks within the Indochina block

The most striking observation is that the magnitude of clockwise rotation estimated from our Eocene data ($90.7 \pm 15.4°$) is significantly larger than those estimated based on Cretaceous paleomagnetic data [8,9] in the northern part of the Simao Terrane. Cretaceous paleomagnetic data from Yunlong [8] and Yongping [9] show clockwise rotation of about 36–38° with respect to the western part of the Yangtze craton.

We can explain a difference in rotation magnitude between Cretaceous and Eocene strata by the superimposition of regional and local rotations. In total, the entire Lanping and Yunlong regions experienced about 40° clockwise rotation until the Eocene as a relatively rigid body rotation of the northern part of the Simao Terrane. The sampling areas in Eocene strata would have further experienced additional clockwise rotation of about 50° independently from the Cretaceous strata sampled elsewhere by the movement of the local faults since the Eocene. Because the Lanping and Yunlong areas are surrounded by several thrusts and strike–slip faults (Fig. 1B,C), they may form as tectonically independent small blocks.

Similar differential rotation was observed at the Jinggu area in the central part of the Simao Terrane which was subjected to clockwise rotational motion greater than its northern part [10,11]. A large declination value ($D = 116°$) is reported from the paleomagnetic direction of the mid-Cretaceous Mangang Formation strata at a locality of 23.5°N/100.7°E, whereas declination of about 80° is found in other areas of the lower Cretaceous Jingxing Formation strata ($D = 85°$) and the mid-Cretaceous Mangang Formation strata ($D = 79°$) as well as the Eocene–Oligocene Mengla Group strata ($D = 85°$). The large declination of the Mangang Formation strata was ascribed to differential rotation of a small block on which the sampling locality is located [11].

An alternative explanation requires counterclockwise rotation of the northern part of the Simao Terrane including the Lanping basin with respect to its central part. The following scenario can be proposed: (A) the northern part independently rotated counterclockwise through about 50° in Cretaceous time, and then (B) about 90° clockwise rotation of a whole Simao block, consisting of its northern and central parts, occurred later than the Late Eocene. This model explains a large amount of rotation of about 90° seen in the central part of the Simao Terrane [11] as well as declination values of the Cretaceous and Eocene strata in the Lanping basin. Unfortunately there is no geologic evidence to support the Cretaceous counterclockwise rotation process of the northern part of the Simao Terrane [34]. Because only clockwise rotation is predicted for deformed terranes situated northeast of India from the laboratory experiments [3,4], it is hard to validate a kinematic perspective of any model involving counterclockwise rotation of the northern part of the Simao Terrane.

We conclude that the discrepancy in rotation between Eocene and Cretaceous paleomagnetic data is a reflection of differential rotation of small blocks within the Simao Terrane. Differential rotation of fault blocks exposing Eocene strata, due to fault movement, may have increased the magnitude of clockwise rotation with respect to the Cretaceous sites in the Lanping basin.

4.3. Southward displacement of the Indochina block

Latitudinal difference between the Lanping basin and the Ya'an area in the Late Eocene can be evaluated from our paleomagnetic data. The paleolatitude of the Lanping basin, calculated from the observed inclination is $22.6 \pm 10.9°$N, whereas the paleolatitude of the Lanping basin expected

from the Ya'an area is 14.0 ± 3.6°N. The latitudinal difference between the observed paleolatitude and the expected paleolatitude from the Ya'an area is 8.6 ± 11.5°, corresponding to 950 ± 1260 km (Fig. 7B).

Southward displacement of the Indochina block has been inferred [8,35,36] from Cretaceous and Jurassic paleomagnetic results. Sato et al. [8] demonstrate 11.9 ± 7.5° southward displacement of the Simao Terrane (Fig. 7A), and Yang and Besse [35] suggest 11.5 ± 6.7° for southward displacement of the Indochina block. Yang et al. [36] estimate 8.4 ± 3.0° southward motion of the terranes situated west of the Red River Fault relative to the South China on the basis of paleomagnetic data of the upper Triassic to upper Cretaceous rocks. Although the latitudinal difference of the Lanping basin estimated from our Eocene paleomagnetic data is within uncertainties, the sense and magnitude of the southward motion are consistent with existing Cretaceous and Triassic paleomagnetic data. The present Eocene data infer southward displacement of the Indochina block in the Eocene/post-Eocene time. Based on the extrusion experiments for Indochina [3,4], this southward displacement is associated with southeastward extrusion of Indochina due to indentation of India into Asia (e.g. [12,14–16]). This is consistent with geologic and chronologic evidence. Geologic relationships and chronologic data indicate that 700 ± 200 km of sinistral motion along the Red River fault took place between 35 and 17 Ma [13–16,41]. In order to better define the Cenozoic extrusion history of Indochina based on paleomagnetism, more reliable Tertiary paleomagnetic data are needed from the Simao Terrane and also the Yangtze craton.

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